

1. Meet Scratch!

Scratch is a fun way to learn coding by snapping colourful blocks together. You can make characters move, dance, and talk — all by dragging and dropping. **No typing needed!**

■ Key Facts about Scratch

- Free to use at **scratch.mit.edu**
- Created at MIT for young learners
- Uses colourful drag-and-drop blocks
- Makes coding visual and creative

2. What Is a Loop?

A loop **repeats actions again and again**. Think about your daily routines — many things you do are really loops in real life!

■ Brushing teeth every morning	Repeating each day
■ Walking steps when moving	Repeating each step
■ Alarm going off every weekday	Repeating each weekday
■ Song chorus played many times	Repeating chorus

Why Use Loops in Coding?

- Save time & effort
- Write less code
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- Make better games
- Create animations

3. The Repeat Block

The Repeat block tells the computer to do something **many times**. You pick a number — and the computer does the action that many times. No need to write the same thing again and again!

Part	Description
Number input	Tells Scratch how many times to repeat

Blocks inside	The actions that get repeated each time
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4. How Does the Repeat Block Work?

You pick a number — that's how many times the block repeats. The computer does the action that many times and then stops. **You are in control!**

Step 1	Run the blocks inside
Step 2	Check: is the count done?
Step 3	If NO → repeat again
Step 4	If YES → stop!

■ Each time the loop runs, the computer counts down until it reaches zero!

5. Examples

Example 1: Draw a Square! ■

To draw a square in Scratch, we use the Repeat block **4 times**. Inside it we put: **Move 100 steps**, then **Turn 90 degrees**. The computer repeats this 4 times and draws a perfect square!

The Scratch Code:

```
when ■ clicked
repeat 4
  move 100 steps
  turn ■ 90 degrees
end
```

■ What happens? Each side is 100 steps long. After 4 repeats, all 4 sides are drawn → perfect square!

Example 2: Move Your Sprite ■

Make your sprite move forward 10 steps, **5 times**! Use the Repeat block with the number 5, and put Move 10 Steps inside it.

The Scratch Code:

```
when ■ clicked
repeat 5
  move 10 steps
end
```

Repeat Block	What Happens
repeat 4	The action happens exactly 4 times
repeat 1	The action happens just once

repeat 10

The action happens 10 times!

6. Common Mistakes with Loops

■ Blocks Outside Repeat

Make sure your action blocks are snapped **inside** the Repeat block. If they are outside, they won't repeat!

■ Wrong Number of Repeats

Check the number in your Repeat block. If it's too big or too small, your sprite won't do what you want!

■ Forgetting to Start

Always connect your Repeat block to a **When Flag Clicked** or another event block — otherwise the code won't run.

7. Activities — Your Turn! ■

Activity 1

Use the Repeat block to make your sprite **jump up and down 3 times**.

*Hint: use **Change Y by 50** (up) and **Change Y by -50** (down).*

Activity 2

Create your own loop animation! Choose what your sprite will do — spin? dance? change colour? Decide how many times to repeat. **Make it your own!** ■

My Activity 2 Plan

My sprite will:

Number of repeats:

My sprite's actions:

8. What Did We Learn Today? ■

- Coding tells computers what to do
- Loops repeat actions to save time
- The Repeat block repeats instructions in Scratch
- We used Repeat to draw shapes and move sprites
- Action blocks must be placed **inside** the Repeat block
- You are always in control of how many times it repeats!

■ **Did You Know?** The word "loop" comes from the idea of going around again and again — just like a loop of ribbon has no end!

■ Keep Coding and Exploring! You are doing an amazing job! Every great coder started just like you. What will you create next? ■